

SQL Notes

What is SQL?

SQL stands for **Structured Query Language**.

It is a language created **to communicate with SQL databases** such as:

- MySQL
- **Oracle Database**
- **PostgreSQL**
- SQLite

We generally get **datasets from platforms** like:

- Kaggle
- Hugging Face

What is a Database?

A database is a **storage system where data is kept and managed**.

Databases are mainly divided into two types:

1. SQL Databases (**Relational**)
 - Data is related to each other.
 - Works similarly to Excel tables.
 - Uses rows and columns.
2. NoSQL Databases
 - Data **may not have relationships**.
 - More flexible structure.
 - Often **works like dictionaries or JSON objects**.
 - Data can be **scattered/distributed**.

Examples:

- **MongoDB**
- Redis

SQLite

SQLite is a **lightweight database**.

Good for:

- Mobile applications
- **Small applications**
- Embedded systems

Not **ideal** for:

- **Very** large systems with **millions of users**

Basic SQL Commands

SELECT

Used **to read data from a table**.

SELECT Example

```
SELECT * FROM demo;  
* means "select everything".
```

Selecting Specific Columns

```
SELECT id, hint FROM demo;
```

Only selected columns are returned.

WHERE Condition

```
SELECT * FROM demo WHERE name = 'Import';
```

Used for filtering rows.
String values use single quotes.

LIMIT

```
SELECT * FROM demo LIMIT 10;
```

Useful when tables contain millions of rows.

Comparison Operators

```
SELECT * FROM demo WHERE id > 10;
```

Returns rows where id is greater than 10.

Empty Result

```
SELECT * FROM demo WHERE id = 30;
```

If there is no matching data, SQL returns an empty table.

LIKE Operator

```
SELECT * FROM demo WHERE name LIKE 'S%';
```

S% -> starts with S

```
SELECT * FROM demo WHERE name LIKE '%a';
```

%a -> ends with a

```
SELECT * FROM demo WHERE name LIKE '%an%';
```

contains an

VARCHAR vs TEXT

Both store strings, but capacities are different.

INSERT

```
INSERT INTO demo (name, hint)
VALUES ('Nemo', 'Test');
```

Used to add new data.

UPDATE

```
UPDATE demo
SET name = 'Ergin'
WHERE id = 29;
```

Used to modify existing data.

Be careful:
Multiple rows may have the same name.
Using id is safer.

DELETE

```
DELETE FROM demo WHERE id = 29;
```

Used to remove data.

CREATE TABLE

```
CREATE TABLE IF NOT EXISTS Students (
    id INTEGER PRIMARY KEY,
    name VARCHAR,
    surname VARCHAR
);
```

PRIMARY KEY uniquely identifies each row.

INSERT INTO TABLE

```
INSERT INTO Students(name, surname)
VALUES ('A', 'B');
```

```
INSERT INTO Students(name, surname)
VALUES ('C', 'D');
```

```
INSERT INTO Students(name, surname)
VALUES ('E', 'F');
```

Alias (AS)

```
SELECT name AS n
FROM Students
WHERE name = 'A';
```

AS is used to rename columns temporarily.

Row vs Column

Row -> One record
Column -> One attribute/field

CRUD Operations

```
Create -> INSERT
Read   -> SELECT
Update -> UPDATE
Delete -> DELETE
```

Important Tips

- SQL keywords are usually written in uppercase.
- Always use WHERE carefully in UPDATE and DELETE.
- Forgetting WHERE may affect all rows.

Dangerous query example:

```
DELETE FROM demo;
```

This deletes everything in the table.

VARCHAR vs TEXT Capacity Table

Type	Capacity
VARCHAR	up to 255 characters
TINYTEXT	up to 255 characters
TEXT	up to 65,535 characters
MEDIUMTEXT	up to 16,777,215 characters
LONGTEXT	up to 4,294,967,295 characters